Expt No: 06

Hamming code
Error detection & correction

Aim:

To write hamming code for error detection & correction.

Code:

```
def hamming_encyption(data):
    m= len(data)
    r=0
    while ( 2**r) <=(m+r+1):
            r+=1;
    r-=1
    n= m+r+1
    code= ['0'] *n
    j=0
    for i in range (1,n):
        if (i &(i-1)!=0):
            code[i]= data[j]
            j+=1
    for i in range (r):
        pos= 2**i
        pair=0
        for j in range (1,n):
            if j & pos:
                pair^= int (code[j])
        code[pos]= str(pair)

return ''.join(code[1:])
```

```
def hammingcode_decryption(data,r,m):
    for i in range(r):
        pos = 2**i
        pari = 0
        for j in range(1, r+m+1):
            if j & pos:
                pari ^= int(code[j])
        if (pari != 0):
            if (data[pos] == '0'):
                data[pos] = '1'
            else
                data[pos] = '0'
    return ''.join(data)

data = input("Enter binary data :")
print("Data after encoding from sender side",
                    hamming_encryption)
rdata = input("Enter data received by the receiver")
print(hammingcode_decryption(rdata, r, rdata length))
```

o/p:

Enter the data: 1101
No of redundant bits r: 3
Data transmitted from sender: 1010101
Index you want to change: 6
Data received from receiver: 1010111
After performing error detection & correction 1010101

Result:

Error detection & error correction using hamming code practical successfully